Joint Fusion of Candidate Controls for Cooperative Multi-USV Target Defense

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***Abstract*—Jointly choosing how a fleet blends learned and Boids controls substantially improves cooperative target defense. We represent each vessel's admissible actuator mixtures as candidate motions, predict their pairwise interactions, and select a complete fleet configuration that balances source-policy preferences with physical compatibility. The method preserves the original learned and Boids controllers and their componentwise control hull. With a fully trained six-vessel source, joint fusion succeeds in 795 of 856 development episodes (92.9%) for training seed 42, exceeding ARBoids by 25.4 percentage points. Against independent selection with the same source weights, candidates, and prediction tensor, it gains 8.1 points through 112 rescues and 43 harms. Across five independently trained source policies, joint selection improves every run, with a mean increment of 3.9 points over independent selection. Matched neural replacements, transferred-source comparisons, and retained high-fidelity trials further characterize the contribution of joint candidate decisions. These completed results use development collections; a frozen held-out comparison determines the final generalization scope.**

***Index Terms*—Multi-USV target defense, cooperative control, controller fusion, joint decision making, candidate interaction.**

I. Introduction

A fleet defending a target against a faster intruder must coordinate how its vessels move together. Individually plausible maneuvers can send two defenders through the same narrow space, leave an interception direction uncovered, or disturb an otherwise successful approach. The central decision is how available controls should be combined across the fleet. Target guarding and cooperative pursuit provide the task setting [1], [2], [3]; learned marine control provides flexible individual maneuvers [4], [5], [6].

ARBoids [7] blends a learned SAC controller with a Boids controller using an adaptive coefficient. This preserves a structured coordination prior while allowing task experience to change the maneuver. We retain these two control sources and their original physical execution model. Our contribution concerns the *joint choice of mixtures*: a vessel's preferred coefficient depends on which control candidates its teammates execute. Observing teammate positions alone does not resolve this dependence. We predict the motion associated with concurrent candidate controls and select a complete configuration using the resulting interaction costs.

The proposed joint fusion rule resolves interactions between actuator-level learned–Boids candidates while preserving the original control hull. Its matched independent comparator receives the same candidates and prediction tensor, allowing the experiments to identify the effect of selecting their combination jointly. We additionally compare state-attention and candidate-attention replacements and retain all five independently trained source policies. Paired successes and failures identify the rescues and harms behind the measured gains.

II. Task and Control Model

A. Dynamics and task outcomes

The numerical environment uses the released ARBoids vessel dynamics and task implementation [7]. A defender obeys

$$\dot{\eta}_i = R(\psi_i)\nu_i, \qquad M\dot{\nu}_i + C(\nu_i)\nu_i + D(\nu_i)\nu_i = \tau_i. \tag{1}$$

The integration period is 0.05 s and the command period is 0.2 s. Each defender thruster remains in $[-500, 1000]$ N; the agility parameter scales the intruder's thrust bounds. The protected radius is 15 m, and both capture and defender-collision thresholds are 5 m.

We preserve the source termination order. Intrusion failure occurs when the intruder enters the protected radius *or* its distance to the target becomes smaller than that of every defender. Defender collision is checked next, followed by capture, and then successful defense at the 80-s deadline. The radial-pass condition is therefore included in reported source failures, even when the intruder has not physically crossed the target boundary. We additionally record physical boundary crossings, captures, collisions, and episode duration.

The controller observes current vessel geometry, measured velocity, and the available learned and Boids commands. It does not observe future disturbances or the intruder's future control sequence. Source dynamics, source policies, and the source opponent are shared across matched comparisons. The source APF opponent has an order-dependent obstacle update; canonical feature ordering does not change that simulator behavior.

B. Control sources and admissible mixtures

Let u_i^{L} and u_i^{B} be the current learned and Boids thrust vectors. We use two actuator coefficients,

$$u_i = g_i \odot u_i^{\mathrm{L}} + (1 - g_i) \odot u_i^{\mathrm{B}}, \qquad g_i = (g_{i,1}, g_{i,2}) \in [0,1]^2. \tag{2}$$

Allowing separate port and starboard coefficients gives the same admissible representation to joint and independent selectors. Both coefficients remain within the original componentwise control hull. The source adapter supplies a scalar preference, which is mapped to a bounded center $\bar{g}_i$ and shared by the two actuators. Each actuator chooses from $\{0, \bar{g}_i, 1\}$,

giving nine candidates per vessel. Duplicate candidates are coalesced.

The primary comparisons train the source policies directly with six vessels for one million steps. Joint and independent selection reuse the same selected ARBoids source weights. The source policy's mean teammate embedding permits observations from other fleet sizes. Earlier experiments transferring a three-vessel source to six vessels are reported as a separate supporting comparison.

III. Joint Selection of Fusion Candidates

A. Candidate motion and pairwise interaction

For every candidate g_i^k, a public-state nominal model predicts the vessel trajectory over 3.2 s at 0.1-s prediction steps. Learned and Boids commands are recomputed every 0.2 s on predicted observations. The candidate coefficient is retained over this forecast, and the executed fleet replans at the physical command rate. The predictor evaluates candidate intent while complete closed-loop episodes measure the resulting task performance.

For two candidates, define

$$d_{ij}^{k\ell} = \min_{\tau\in[0,H]} \|\hat{p}_i^k(\tau) - \hat{p}_j^\ell(\tau)\|, \qquad \psi_{ij}(k,\ell) = [7 - d_{ij}^{k\ell}]_+^2. \tag{3}$$

The unary term penalizes departure from the source preference,

$$\phi_i(k) = \frac{1}{2}\sum_{a=1}^{2}\left(\frac{g_{i,a}^k - \bar{g}_i}{0.4}\right)^2. \tag{4}$$

The joint fleet configuration minimizes

$$\boldsymbol{k}^* \approx \arg\min_{\boldsymbol{k}}\left[\sum_i \phi_i(k_i) + 32\sum_{i<j}\psi_{ij}(k_i,k_j)\right]. \tag{5}$$

The interaction term depends on both vessels' selected candidates. Its role is to distinguish complete combinations that can have similar individual preferences but different physical compatibility.

B. Joint search and the matched independent selector

A width-128 beam constructs complete configurations, followed by three coordinate sweeps. The refined selector additionally refines every retained beam endpoint and performs one pair-block sweep. It accepts a refinement only when both total preference departure and total predicted interaction cost are no worse than the original joint solution. This improves search within the same candidate library and objective.

The independent comparator receives the identical candidates, trajectories, and pairwise cost tensor. With source-preference probabilities $p_j(\ell) \propto \exp[-\phi_j(\ell)]$, it chooses

$$k_i^{\mathrm{ind}} = \arg\min_k\left[\phi_i(k) + 32\sum_{j\neq i}\sum_\ell p_j(\ell)\psi_{ij}(k,\ell)\right]. \tag{6}$$

Thus both methods account for the same predicted candidate interactions. The comparison isolates the consequence of selecting a joint configuration instead of minimizing expected cost independently. Joint optimization has additional search cost, which is reported separately from prediction cost.

TABLE I
SIX-VESSEL-SOURCE DEVELOPMENT COMPARISON, TRAINING SEED 42. EVERY METHOD IS EVALUATED ON THE SAME 856 PAIRED EPISODES. JOINT AND INDEPENDENT FUSION SHARE THE SOURCE CHECKPOINT AND CANDIDATE PREDICTIONS.

Method	Source /256	4310 /200	Earlier /200	Later /200	Total /856
SAC	42	27	20	29	118
RP	99	82	80	80	341
ARBoids	171	135	137	135	578
Independent fusion	220	167	166	173	726
Joint fusion	238	188	180	189	795

IV. Experimental Design

The primary comparisons evaluate joint fusion against adequately trained source baselines and against independent fusion with matched candidates and predictions. SAC, residual policy (RP), and ARBoids each receive the original million-step training budget. Five independently initialized source-policy runs use seeds 42–46. Joint and independent fusion share the resulting ARBoids checkpoint within each run. Each baseline run supplies the checkpoint with the highest training-evaluation success rate, breaking ties by mean reward and retaining the first exact tie.

All primary contrasts pair the same initial layout, agility, fleet size, and future seed. Repeated futures from one layout remain in the same bootstrap cluster. Success differences, rescue/harm counts, collision, and capped duration are reported together. Capture-only duration is secondary because successful episodes can differ across methods. Independent source-training repetitions are distinguished from additional evaluation scenes and future samples.

The 856-episode development collection contains 256 episodes from the original source suite and 200 episodes in each of three later suites. The source suite repeats 64 layouts under four futures; the other suites contain new layouts. Repeated use in development and model selection prevents treating these collections as a frozen test. The reserved 4322 layout family remains outside this development comparison.

V. Results

A. Overall performance and independent training runs

Joint fusion succeeds in 795 of 856 episodes (92.9%) with the six-vessel source trained at seed 42, compared with 578 (67.5%) for ARBoids and 341 (39.8%) for RP. Its gain over ARBoids is 25.4 percentage points. It records 793 captures, nine collisions, and 52 source-defined failures; two successful defenses reach the deadline. ARBoids records 578 captures, 270 collisions, and eight source-defined failures. The reduction in collisions therefore accompanies an increase in source-defined failures, which remains part of the comparison.

All 15 source-training runs complete one million steps and the full 856-episode comparison. Table II reports every training seed. Across the five runs, joint fusion achieves $81.6 \pm 19.3\%$ success, independent fusion $77.7 \pm 18.5\%$, and ARBoids $65.6 \pm 14.2\%$ (mean and sample standard deviation). The 92.9% result conditions on seed 42; the five-run results retain the substantial source-policy variability.

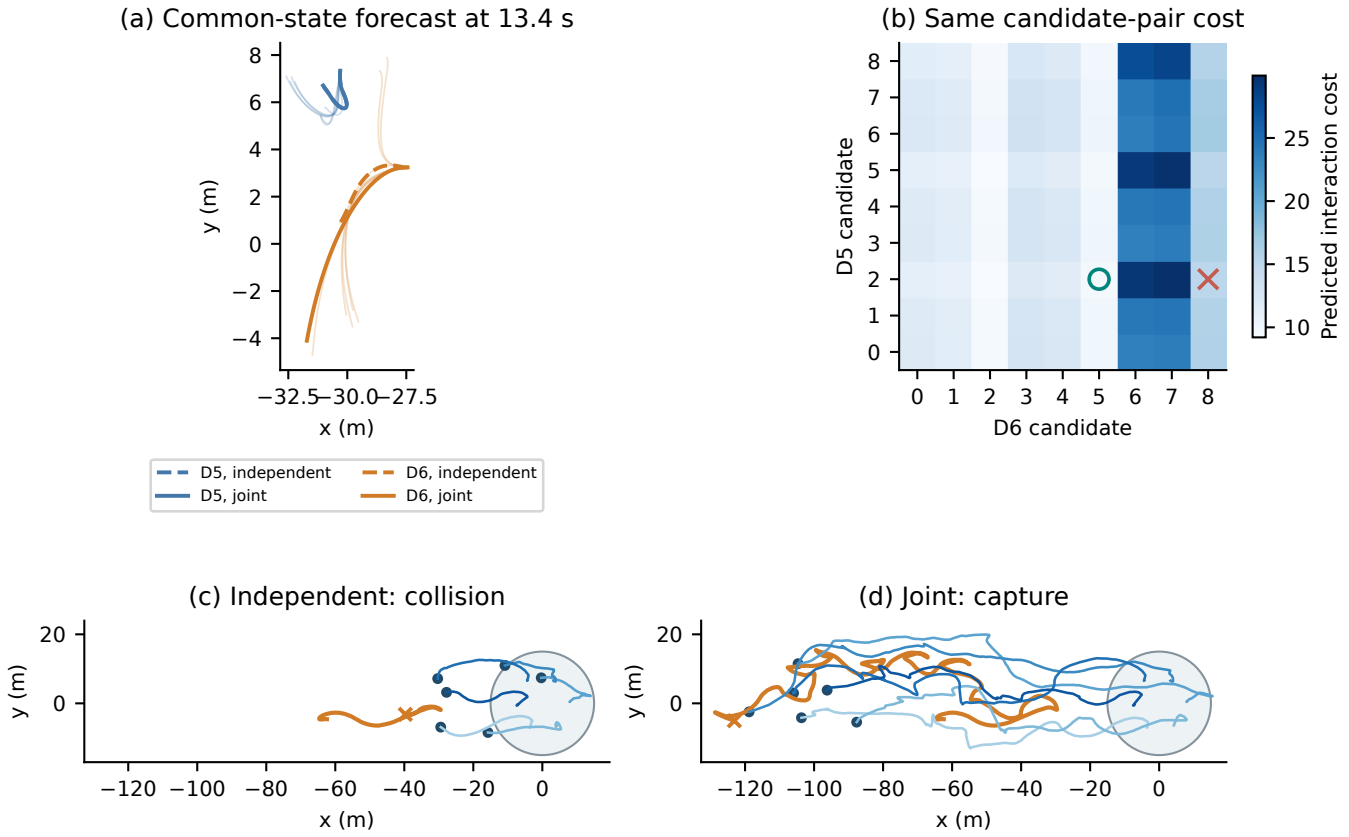


Fig. 1. Joint selection with the fully trained six-vessel source on development scene 431020053. The top panels compare identical candidate forecasts at a common state at 13.4 s on the independent trajectory, where the selected pair has its largest predicted risk reduction. The complete joint objective decreases from 495.5 to 339.0. The bottom panels replay both policies from the same initial state and future: independent selection collides at 13.6 s, whereas joint selection captures at 70.6 s. This illustration is the first joint rescue of an independent collision in the fixed 4310 row order.

TABLE II
FIVE INDEPENDENT SIX-VESSEL SOURCE-TRAINING RUNS AND THEIR PAIRED FUSION SELECTORS. EACH CELL COUNTS SUCCESSFUL DEFENSES IN THE SAME 856 DEVELOPMENT EPISODES. JOINT AND INDEPENDENT FUSION REUSE THE ARBOIDS CHECKPOINT IN THEIR ROW.

Seed	SAC	RP	ARBoids	Independent	Joint
42	118	341	578	726	795
43	212	337	355	385	404
44	182	253	584	773	780
45	320	247	615	736	758
46	145	252	675	707	755

B. Direct evidence for joint selection

With the fully trained six-vessel ARBoids source at seed 42, joint selection raises success from the independent selector's 726/856 (84.8%) to 795/856 (92.9%). The gain is 8.1 percentage points, with a layout-cluster bootstrap interval of [5.3,11.1] points, comprising 112 rescues and 43 harms. Captures increase from 723 to 793, collisions decrease from 20 to nine, and source-defined failures decrease from 110 to 52. These development intervals describe resampling uncertainty and do not correct for adaptive model selection.

The per-seed joint-selection gains are 8.1, 2.2, 0.8, 2.6, and 5.6 points, averaging 3.9 ± 2.9 points. Joint selection improves every run, while the magnitude varies with the learned source. Because the two selectors share source weights, candidate predictions, and refinement settings, this comparison directly measures the effect of resolving candidate combinations jointly.

On the same six-vessel source and 856 episodes, state-attention and candidate-attention replacements obtain 626 (73.1%) and 665 (77.7%) successes. Both use two attention layers of width 128, 192,769 parameters, 24,000 updates, and the same 24,576 measured teacher queries from teacher and student trajectories. The joint selector provides training supervision; each replacement makes its own neural choices at deployment. Conditioning each candidate on only the teammates' source-preferred reference trajectories yields 754 successes (88.1%), compared with 726 for independent expectation over teammate candidates and 795 for joint selection.

C. Retained high-fidelity comparison

The transferred-source VRX comparison [8] uses two settings with agility 2.5 and 3.5 and two scene seeds per condition, yielding eight paired scenes per method. VRX uses a 0.2-s control period, 15-m target radius, 5-m collision radius, 5.5-m capture radius, and 100-s time limit. These historical deployment settings are distinct from the numerical 80-s task.

The standalone joint predecessor captures in all eight scenes. Refined joint fusion captures in seven and collides in one, independent selection captures in four, and the transferred source captures in two. The joint versions retain their separate identities: the numerical refinement does not improve every high-fidelity scene. This small development batch supports the paired deployment comparison without replacing the numerical repetition study.

VI. Discussion

Joint fusion makes the dependence between candidate decisions explicit. Its pairwise costs distinguish configurations that share plausible individual preferences but induce different fleet motion. The matched independent selector uses the same physical predictions, so its comparison with joint selection is more specific than an overall comparison with the source controller.

The interaction objective is a soft risk cost evaluated with a nominal predictor. Source-policy preferences remain influential, and a low predicted cost does not guarantee collision-free execution or capture. Additional joint search also consumes computation beyond candidate prediction. The observed reliability gains, source-defined failures, paired harms, and variation across training seeds jointly delimit the supported conclusion. Current results describe completed development comparisons; held-out evaluation must use a frozen method and comparators.

VII. Conclusion

Joint fusion improves cooperative target defense by selecting compatible fleet mixtures from common learned and Boids controls. The actuator-level representation and pairwise motion model allow the selector to resolve candidate interactions while preserving the source control hull. With a fully trained six-vessel source at seed 42, the method achieves 92.9% success and gains 8.1 points over matched independent selection. Joint selection improves all five source-training runs, averaging a 3.9-point increment. The transferred-source comparison additionally yields a 10.7-point refined joint-selection gain, while the high-fidelity outcomes retain both successful defenses and a refinement failure.

Appendix A
Transferred-Source Joint-Fusion Evidence

These completed comparisons use a source trained with three defenders and transferred to six. They remain distinct from the fully trained six-vessel-source comparison in the main text.

A. Joint selection and neural replacements

The refined joint selector obtains 92 more successes than the independent selector across the same 856 episodes, an absolute gain of 10.7 points (development layout-cluster interval [7.5,14.1]). It rescues 144 independent failures while harming 52 independent successes. Its per-suite increments are 12.1, 11.5, 9.5, and 9.5 points. Because both methods receive the same nine candidates per vessel and the same prediction tensor, this contrast directly measures the value of resolving their combination jointly.

TABLE III
Transferred-source development comparisons on 856 common episodes. Joint and refined joint selection remain separate configurations.

Method	Source /256	4310 /200	Earlier /200	Later /200	Total /856
Source	37	21	14	25	97
Independent fusion	205	153	153	160	671
Joint fusion	225	171	169	173	738
Refined joint fusion	236	176	172	179	763

Joint selection increases captures from 669 to 762, reduces defender collisions from 114 to 63, and reduces source-defined intrusion failures from 71 to 30. The collision-rate reduction is 6.0 points [3.3,8.8]. On the 616 jointly captured episodes, its paired mean capture time changes by -0.24 s $[-1.27,0.77]$.

The original source backbone and four independently trained source backbones give five separate joint-versus-independent comparisons on the common 228-episode screen. Joint selection obtains 948/1,140 successes, compared with 843/1,140 for independent selection. The mean gain is 9.2 points; individual gains are 12.3, 4.4, 7.9, 10.1, and 11.4 points. All four additional backbones complete the original one-million-step source training. This supporting experiment repeats the analytic joint selector across independently learned source commands on its own 228-episode development screen.

The strengthened state-attention and candidate-attention alternatives obtain 446/856 (52.1%) and 551/856 (64.4%) successes. Both use two attention layers of width 128, 192,769 trainable parameters, 24,000 optimizer updates, and the same 24,576 measured teacher queries collected on teacher and student trajectories. Training balances changed and unchanged teacher choices and retains every physically equivalent candidate label; candidate trajectories are expressed in a common task frame. The joint selector supplies supervision only during training. At deployment, each replacement selects candidates through its own neural outputs. These fitted alternatives improve substantially over their initial versions but remain below independent selection (78.4%) and joint selection (89.1%). The matched joint-versus-independent contrast provides the direct evidence for joint decision making; the replacement comparison additionally tests whether these neural representations recover its benefit.

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